

Part B: Direct AS1–AS4 Peering and Delay Comparison

COL7524 – Advanced Computer Networks

Objective. We implemented a policy in the Part B topology where AS1 prefers to send traffic through AS2 and AS3, even though a direct AS1→AS4 link is faster. The policy is implemented via two mechanisms: a static route and an interface-metric adjustment in ns-3's global routing.

1. Part B Configuration

A separate `bgp_partB.cc` file was created from the provided `bgp.cc` file. A direct point-to-point link was added between AS1 and AS4 with a data rate of 10 Mbps and a propagation delay of 2 ms. The original links have delays of 3 ms, 4 ms, and 3 ms, so the new link has a lower delay than any of the links in the original chain.

Available paths: AS1 → AS2 → AS3 → AS4 (3 hops) and AS1 → AS4 (1 hop).

Observed path: The network selected the direct one-hop path AS1 → AS4. The first packet was received through the 192.168.4.x subnet assigned to the new direct link, confirming that traffic did not traverse AS2 and AS3.

2. Routing Path and Reason

The direct AS1 → AS4 path is one hop, while the original route is 3 hops. After adding the extra link, the direct path is preferred because ns-3 global routing selects the lower-cost route. The direct link also has a smaller propagation delay than either of the original links.

3. Delay Calculation

Part A had an average end-to-end delay of 11.8066625 ms. The generated FlowMonitor XML report for Part B shows `delaySum = 1.94688 × 107 ns` and `rxPackets = 8` for the forward flow.

$$\begin{aligned}\text{Average delay} &= \text{delaySum} \div \text{rxPackets} \\ &= (1.94688 \times 10^7 \text{ ns}) \div 8 \\ &= 2.4336 \times 10^6 \text{ ns} \\ &\approx 2.434 \text{ ms}\end{aligned}$$

4. Delay Comparison

Metric	Value
Part A average end-to-end delay	11.807 ms
Part B average end-to-end delay	2.434 ms
Absolute reduction	9.373 ms

$$\begin{aligned}\text{Percentage reduction} &= ((11.8066625 - 2.4336) / 11.8066625) \times 100 \\ &\approx 79.40\%\end{aligned}$$

The direct one-hop path AS1 → AS4 reduces the average end-to-end delay from 11.807 ms to 2.434 ms (an absolute reduction of about 9.373 ms, which is a 79.40% reduction). This reduction is consistent with replacing the original three-link path with a single, lower-delay link.

5. Conclusion

After the direct AS1→AS4 peering link is added, the network selects the one-hop path AS1 → AS4 instead of the path AS1 → AS2 → AS3 → AS4. - Average end-to-end delay Part A baseline = 11.807 ms. Part B = 2.434 ms. Percentage improvement = $(11.807 - 2.434) / 11.807 = 79.40\%$. All 8 packets that were sent were received successfully with no loss.